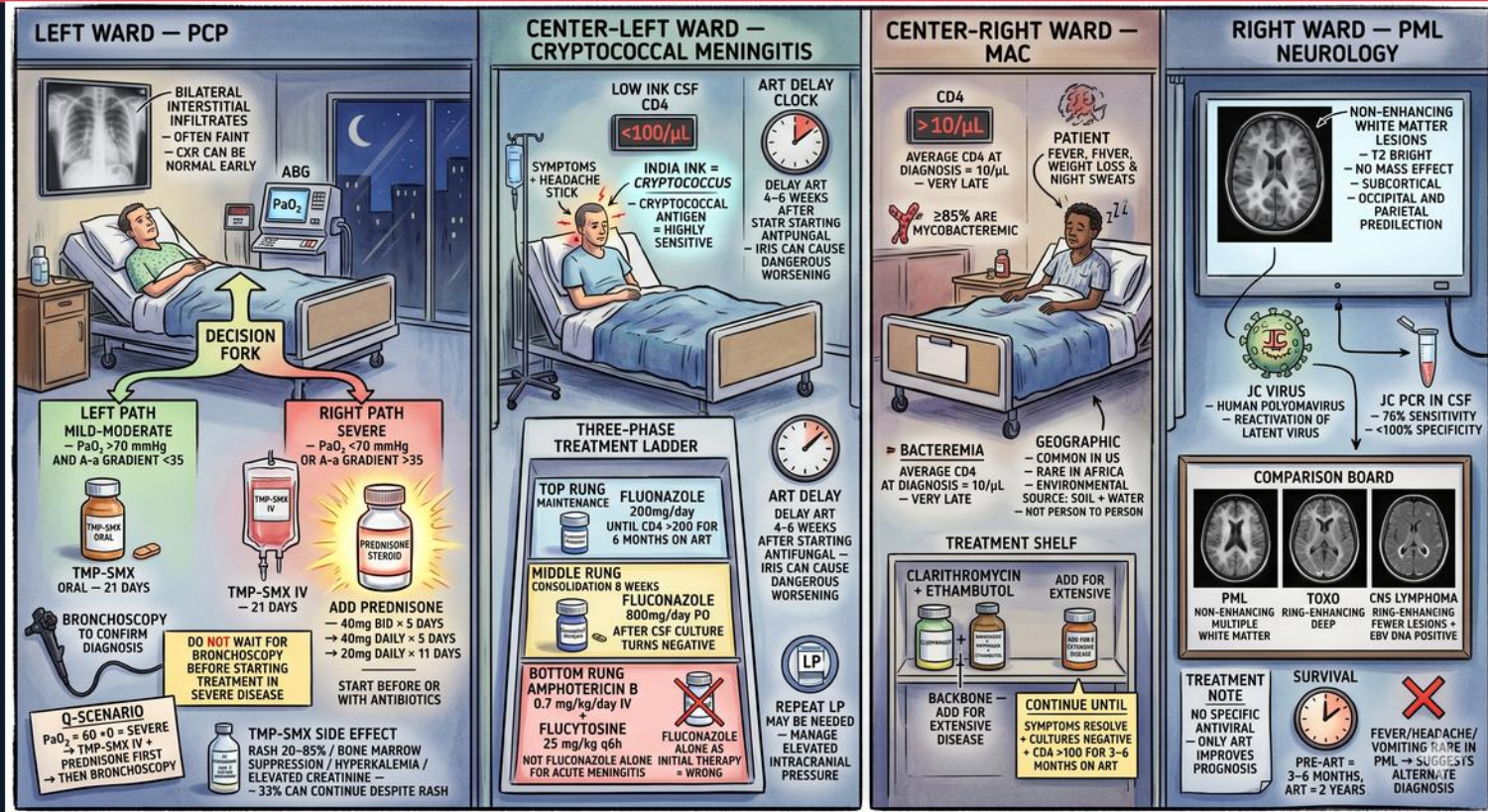


HIV — OI Treatment Wards (PCP, Cryptococcus, MAC, PML)



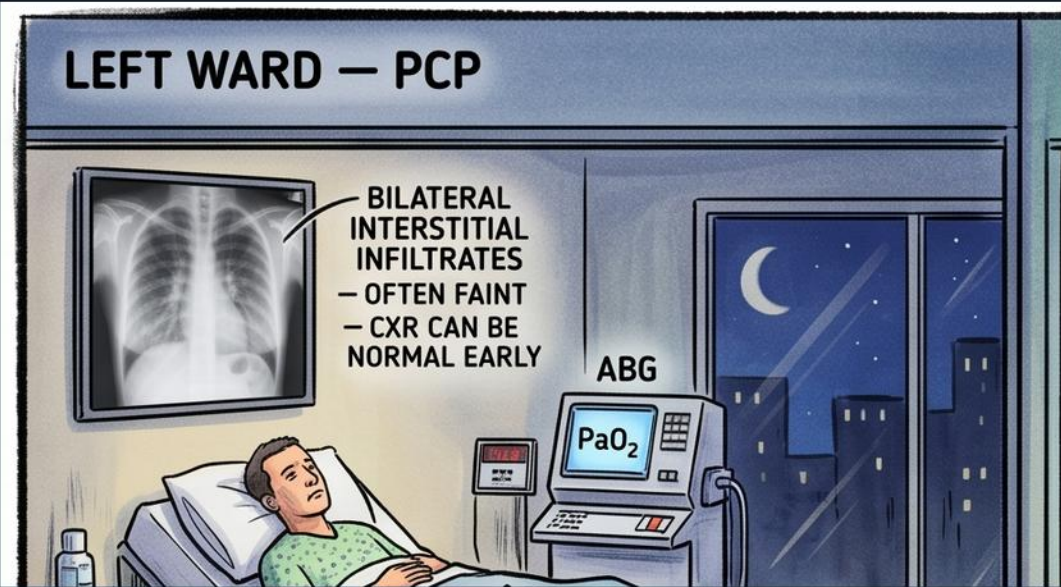
LEFT WARD — PCP



BILATERAL
INTERSTITIAL
INFILTRATES
— OFTEN FAINT
— CXR CAN BE
NORMAL EARLY

ABG

PaO₂



PCP ward / CXR

WHAT YOU SEE

Left ward 'PCP', chest film with bilateral interstitial/ground-glass infiltrates

WHAT IT ENCODES

Pneumocystis jirovecii pneumonia (PCP/PJP) classically shows diffuse bilateral perihilar interstitial (ground-glass) infiltrates, but the CXR can be normal early in disease. It presents subacutely over days to weeks with dry cough, progressive exertional dyspnea, fever, and profound hypoxemia out of proportion to imaging. Risk rises sharply at CD4 <200. LDH is often elevated and beta-D-glucan is a useful serum adjunct; definitive diagnosis is by induced sputum or BAL with silver/immunofluorescent stain showing cysts.

THE BOARD TRAP

WRONG: 'Normal chest X-ray rules out PCP.' A normal CXR does NOT exclude PCP — up to ~25% are normal early. The discriminator is an elevated A-a gradient/exercise desaturation and high LDH; if suspicion is high, get HRCT (shows ground-glass) and induced sputum/BAL rather than stopping the workup.

ABG / A-a gradient fork

WHAT YOU SEE

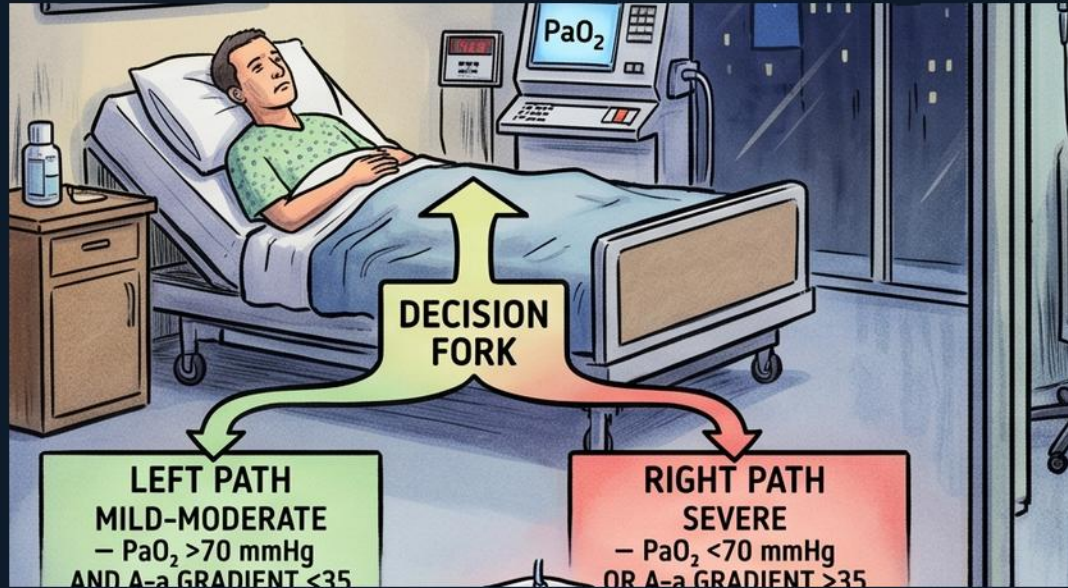
ABG and 'decision fork' for severity

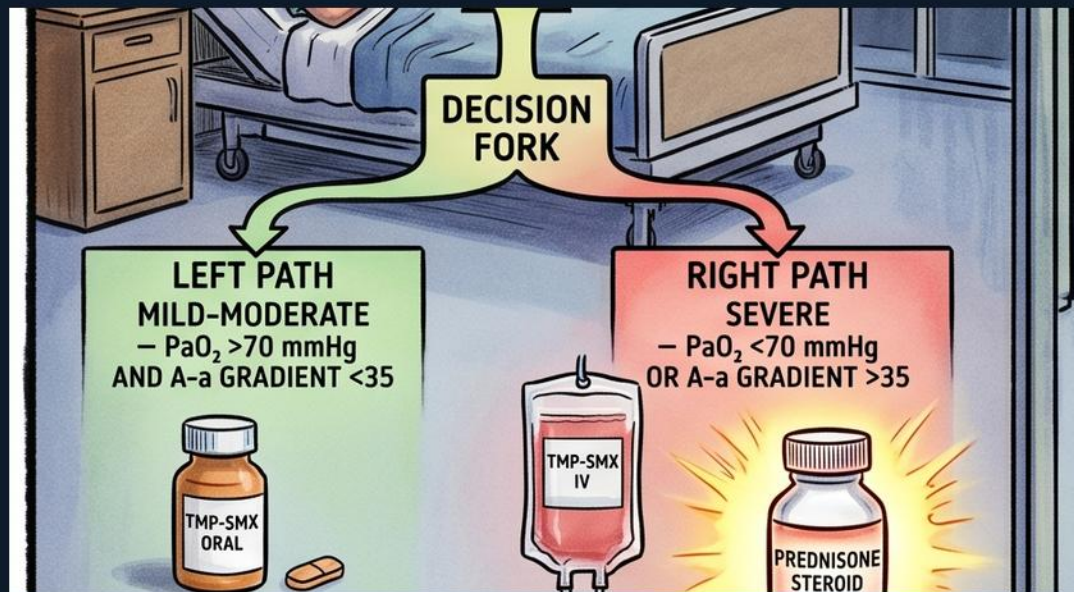
WHAT IT ENCODES

Severity in PCP is stratified by room-air ABG: calculate PaO_2 and the alveolar-arterial (A-a) oxygen gradient to decide whether adjunctive corticosteroids are indicated. The two thresholds to memorize are $\text{PaO}_2 < 70$ mmHg OR A-a gradient > 35 mmHg, which define severe disease. This single decision drives whether you add steroids before/with antibiotics.

THE BOARD TRAP

WRONG: 'Use pulse-ox saturation to decide on steroids.' The board answer requires an ABG-derived PaO_2 and A-a gradient, not SpO_2 . A patient with a borderline saturation but $\text{PaO}_2 < 70$ or A-a > 35 on ABG still meets criteria for steroids — failing to obtain the ABG misses the indication.





Mild PCP path

WHAT YOU SEE

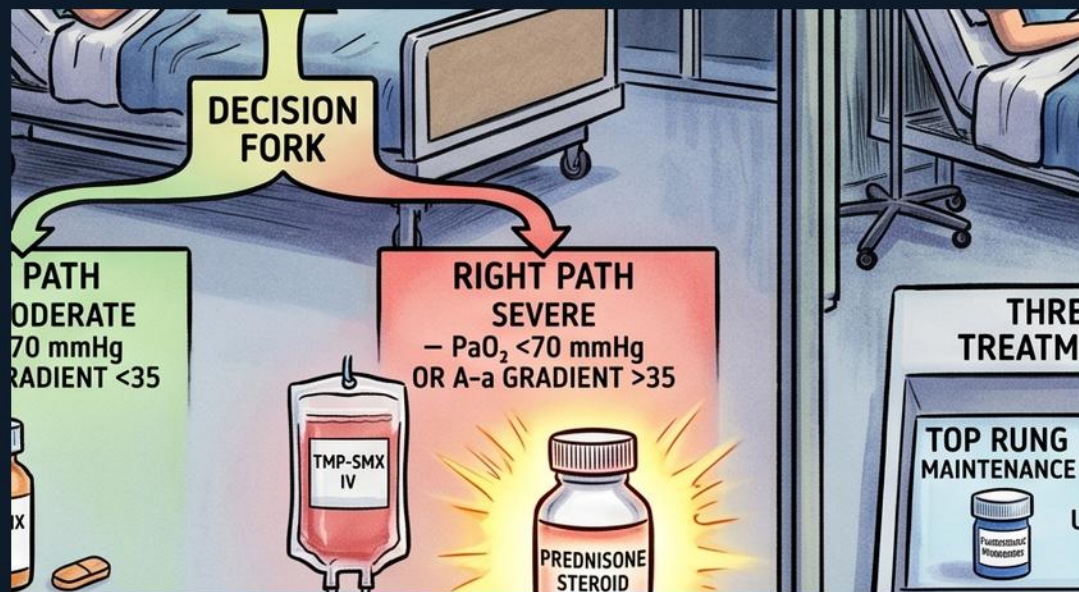
Left path: mild-moderate, $\text{PaO}_2 > 70$ and A-a $< 35 \rightarrow$ TMP-SMX 21 days

WHAT IT ENCODES

Mild-to-moderate PCP ($\text{PaO}_2 > 70$ mmHg AND A-a gradient < 35 mmHg): treat with TMP-SMX for 21 days, NO adjunctive steroids. Dose is TMP 15–20 mg/kg/day (of the TMP component) divided q6–8h, IV or oral depending on severity. After completing treatment, continue secondary prophylaxis (lower-dose TMP-SMX) until CD4 > 200 for ≥ 3 months on ART.

THE BOARD TRAP

WRONG: 'Give steroids to everyone with PCP to be safe.' Steroids are reserved for severe disease ($\text{PaO}_2 < 70$ or A-a > 35); adding them to mild PCP exposes the patient to needless immunosuppression and worsens other untreated coinfections without benefit. Match the therapy to the ABG-defined severity.



Severe PCP + steroids

WHAT YOU SEE

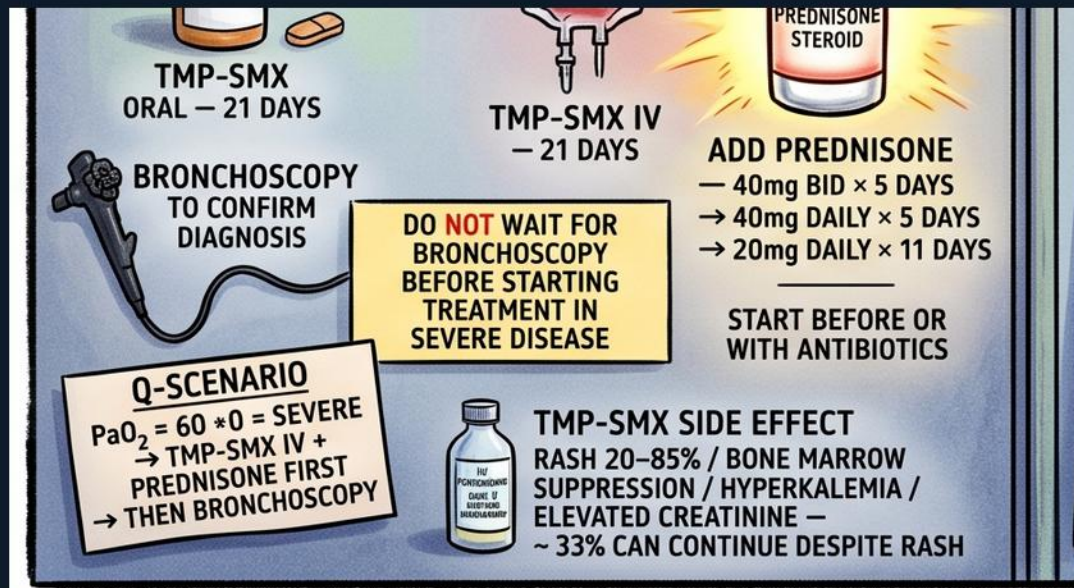
Right path: severe PaO₂ <70 or A-a >35 → ADD prednisone before/with antibiotics

WHAT IT ENCODES

Severe PCP (PaO₂ <70 mmHg OR A-a gradient >35 mmHg): add corticosteroids, started BEFORE or WITH the first antibiotic dose to blunt the inflammatory burst from dying organisms. Standard regimen is prednisone 40 mg BID x5 days, then 40 mg daily x5 days, then 20 mg daily x11 days (taper over 21 days). Adjunctive steroids reduce mortality and respiratory failure in severe PCP.

THE BOARD TRAP

WRONG: 'Start antibiotics first, then add steroids once the patient deteriorates.' Late steroids do not help — they must be given before or within 72h (ideally with the first antibiotic dose) because their benefit is preventing the early inflammatory worsening that occurs as organisms lyse. Delaying steroids until decompensation misses the window.



TMP-SMX is first-line

WHAT YOU SEE

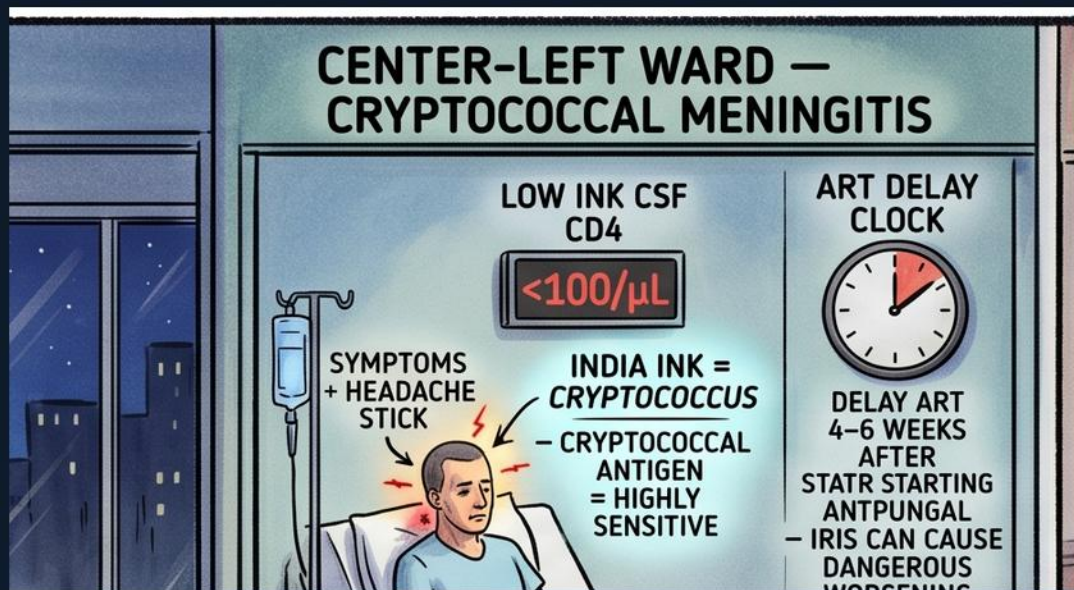
TMP-SMX 21 days bottle

WHAT IT ENCODES

TMP-SMX (trimethoprim-sulfamethoxazole) is first-line for both PCP treatment (21 days) and prophylaxis (one DS or SS tablet daily). Common adverse effects include rash, fever, hyperkalemia, cytopenias, and a benign creatinine rise (trimethoprim blocks tubular creatinine secretion). True alternatives for sulfa allergy/intolerance: primaquine + clindamycin, atovaquone (mild disease), or IV pentamidine (severe).

THE BOARD TRAP

WRONG: 'Any rash on TMP-SMX means stop the drug and switch.' Mild rashes or fever can often be treated through, and many patients tolerate continuation; reserve switching for true severe hypersensitivity (Stevens-Johnson, anaphylaxis), severe cytopenias, or hepatotoxicity. Check G6PD before using primaquine or dapsone-containing alternatives.



Cryptococcal meningitis

WHAT YOU SEE

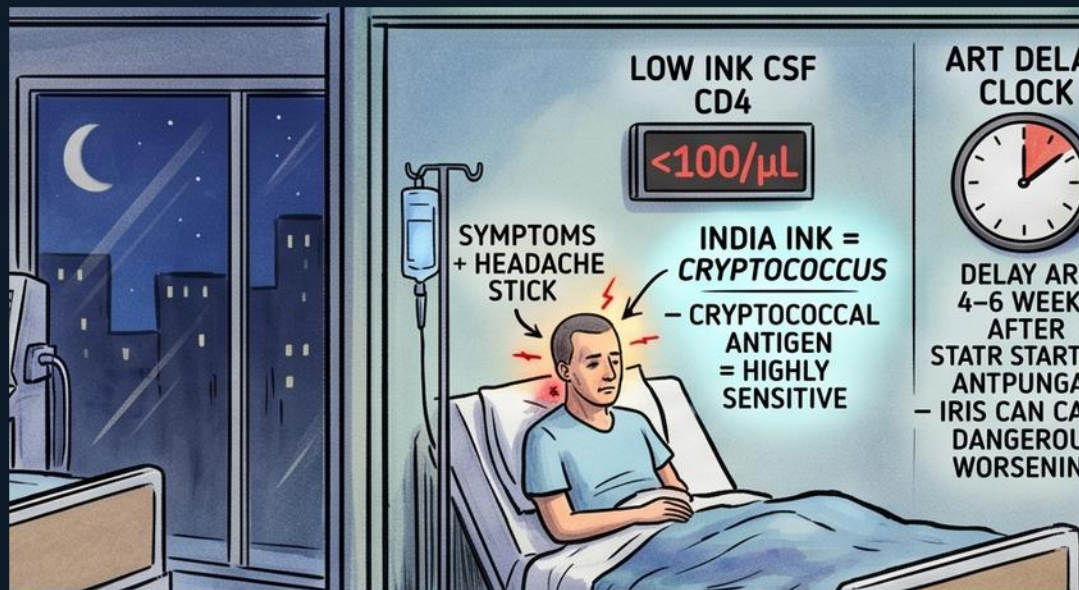
Center-left ward, patient with headache, 'Cryptococcal Meningitis'

WHAT IT ENCODES

Cryptococcal meningitis (*Cryptococcus neoformans*) presents subacutely over days to weeks with headache, fever, malaise, and often subtle meningismus; it occurs typically at CD4 <100. Photophobia and frank nuchal rigidity may be absent. The key complication driving morbidity/mortality is raised intracranial pressure. Diagnose with serum and CSF cryptococcal antigen (CrAg, highly sensitive/specific), India ink, and fungal culture.

THE BOARD TRAP

WRONG: 'Absence of neck stiffness rules out cryptococcal meningitis.' In advanced HIV the inflammatory response is blunted, so classic meningeal signs are frequently absent — a subacute headache with fever at low CD4 warrants LP with CrAg and opening-pressure measurement regardless of exam. Relying on meningismus to exclude it misses cases.



India ink / CrAg + ICP

WHAT YOU SEE

'Low India ink CSF <100 μ L', high opening pressure clock

WHAT IT ENCODES

Diagnosis: CSF/serum cryptococcal antigen (CrAg) is the most sensitive test; India ink shows encapsulated yeast in CSF. CSF typically shows low cell count, low glucose, high protein. Measure opening pressure at every LP — pressures are often markedly elevated (>25 cm H₂O). Manage raised ICP with serial therapeutic lumbar punctures, draining CSF to reduce pressure (and a lumbar drain or shunt if refractory).

THE BOARD TRAP

WRONG: 'Treat elevated ICP in cryptococcal meningitis with mannitol/acetazolamide/steroids.' Pharmacologic measures and steroids do NOT help and steroids increase harm; the proven intervention is repeated therapeutic LPs to mechanically drain CSF. Failure to aggressively manage ICP is a leading cause of death even when antifungals are working.



3-phase tx ladder

WHAT YOU SEE

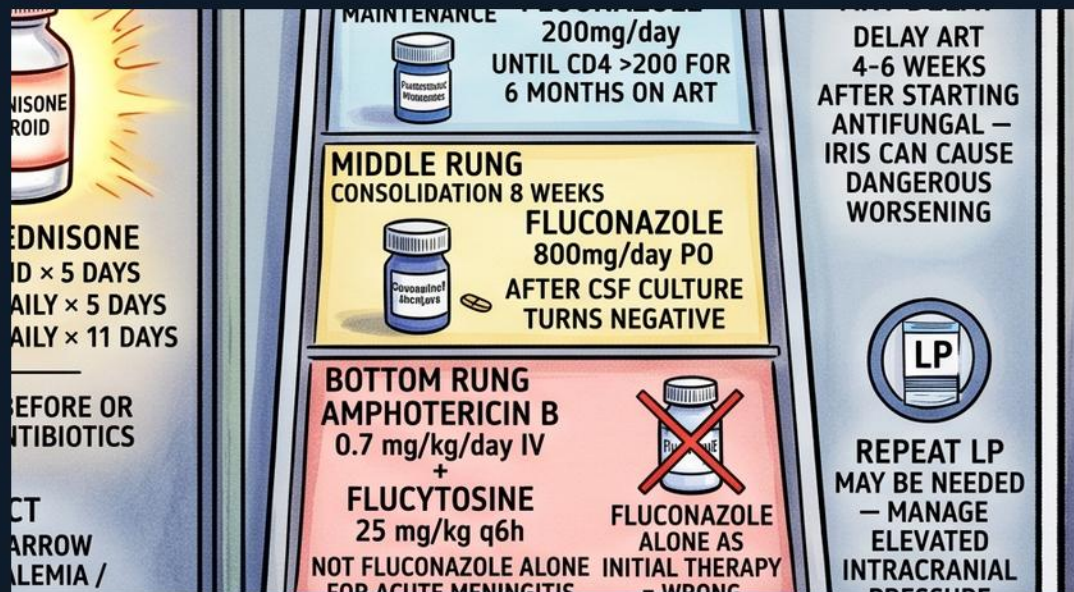
'Three-phase treatment ladder': induction, consolidation, maintenance

WHAT IT ENCODES

Cryptococcal meningitis is treated in three sequential phases: INDUCTION (liposomal amphotericin B + flucytosine, ~2 weeks), CONSOLIDATION (fluconazole 400–800 mg/day, ~8 weeks), then MAINTENANCE/secondary prophylaxis (fluconazole 200 mg/day). Maintenance continues until CD4 >100 for ≥3 months with suppressed viral load on ART. Each step has a defined drug and duration.

THE BOARD TRAP

WRONG: 'Once the patient improves on induction, switch straight to maintenance-dose fluconazole.' You cannot skip the 8-week consolidation phase of higher-dose fluconazole (400–800 mg); jumping to maintenance dosing too early causes relapse. The phases and their durations are sequential and must each be completed.



Ampho B + flucytosine

WHAT YOU SEE

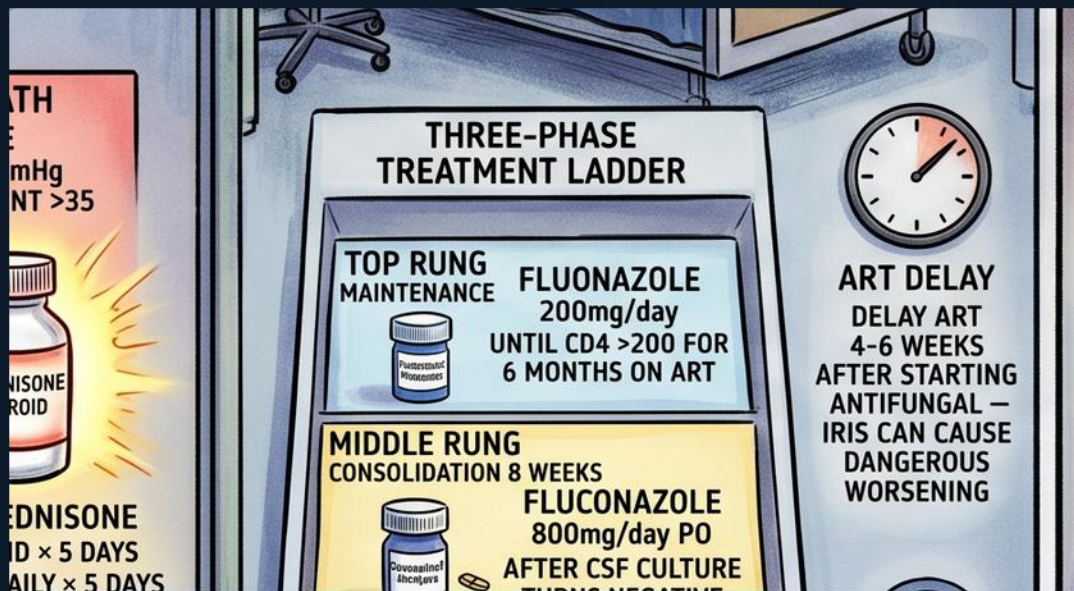
Bottom rung: Amphotericin B 0.7 mg/kg/day + flucytosine (induction)

WHAT IT ENCODES

INDUCTION therapy = liposomal amphotericin B (3–4 mg/kg/day; or deoxycholate 0.7–1 mg/kg/day) PLUS flucytosine (5-FC, 100 mg/kg/day divided q6h) for ~2 weeks. The combination clears CSF faster and lowers mortality versus either alone. Monitor renal function/electrolytes (ampho: K⁺, Mg²⁺ wasting) and flucytosine levels/CBC (marrow suppression).

THE BOARD TRAP

WRONG: 'Fluconazole monotherapy is adequate induction for cryptococcal meningitis.' Fluconazole alone is NOT sufficient induction — the standard is amphotericin B PLUS flucytosine. Fluconazole only enters during the consolidation phase. Choosing fluconazole induction is a classic wrong answer that increases mortality.



Fluconazole maintenance

WHAT YOU SEE

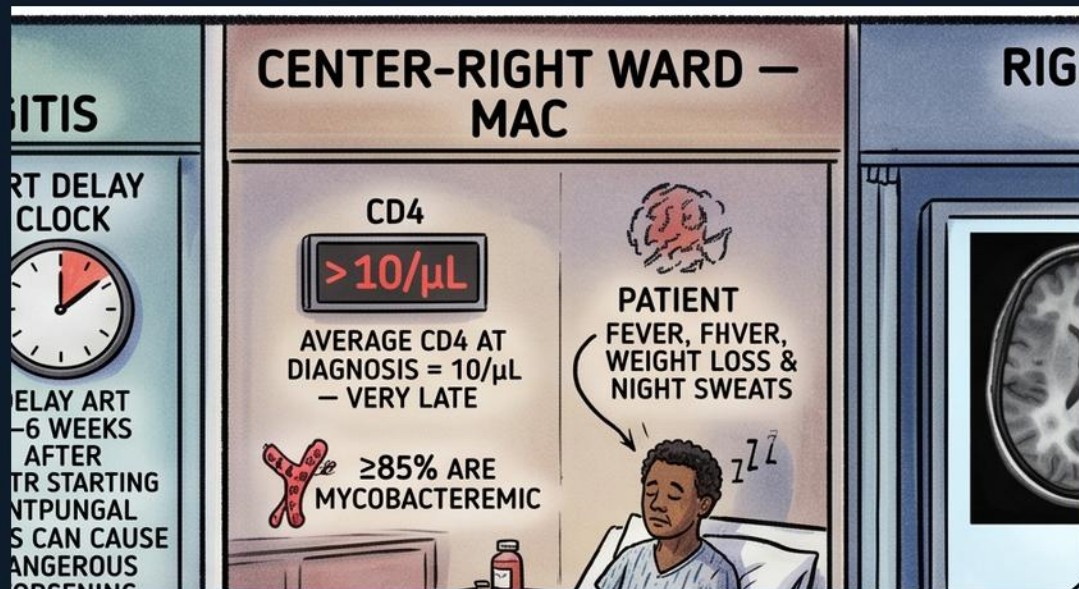
Top/middle rungs fluconazole consolidation then maintenance

WHAT IT ENCODES

After ~2 weeks of induction (and documented clinical improvement/CSF sterilization), step down to fluconazole: 400–800 mg/day for ~8 weeks (consolidation), then 200 mg/day (maintenance/secondary prophylaxis). Continue maintenance until CD4 >100 sustained for ≥3 months on suppressive ART. ART is typically delayed ~4–6 weeks after starting antifungals to reduce IRIS-related death.

THE BOARD TRAP

WRONG: 'Start ART immediately along with antifungals in cryptococcal meningitis.' Unlike most OIs, you **DELAY** ART by ~4–6 weeks here because early ART triggers IRIS that markedly increases mortality. Starting ART concurrently with antifungal induction is the trap — this is the major exception to 'treat all/start ART promptly.'



MAC ward

WHAT YOU SEE

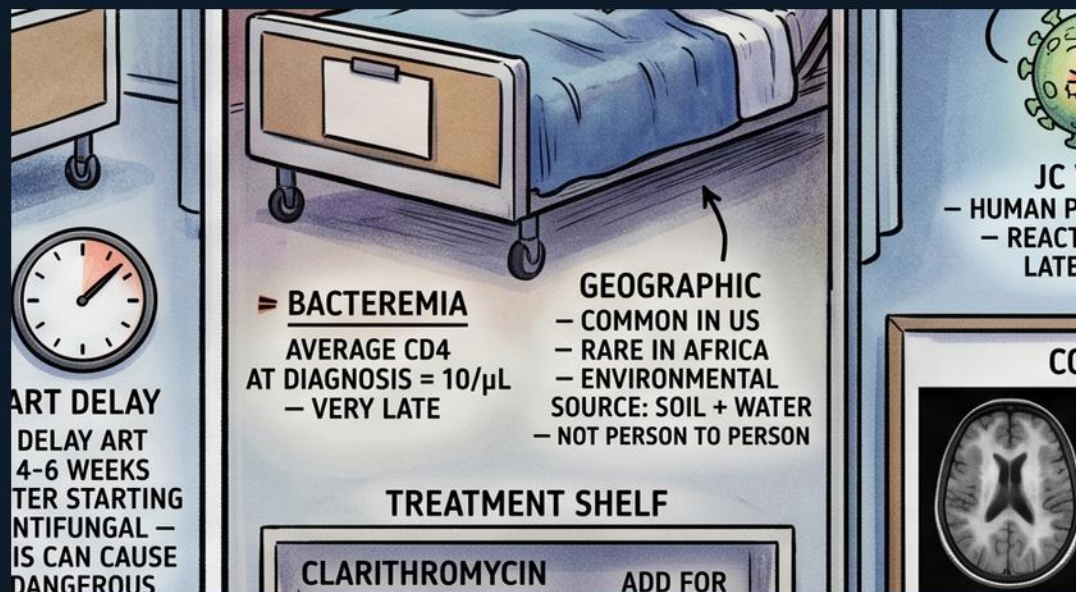
Center-right ward 'MAC', CD4 $<10/<50$, fever, night sweats

WHAT IT ENCODES

Disseminated Mycobacterium avium complex (MAC) is a very late OI, typically at CD4 <50 (often ~ 10). Presents with persistent fever, drenching night sweats, weight loss, fatigue, diarrhea, hepatosplenomegaly, and lymphadenopathy. Labs show anemia and elevated alkaline phosphatase; diagnosis is by mycobacterial blood cultures (or bone marrow/tissue culture).

THE BOARD TRAP

WRONG: 'Fever, night sweats, and weight loss at low CD4 must be tuberculosis.' MAC and TB overlap clinically, but MAC is the very-late (CD4 <50), blood-culture-positive, disseminated entity with high alk phos and hepatosplenomegaly; TB can occur at any CD4 and is more pulmonary. Mycobacterial blood cultures distinguish them — don't anchor on TB.



MAC treatment combo

WHAT YOU SEE

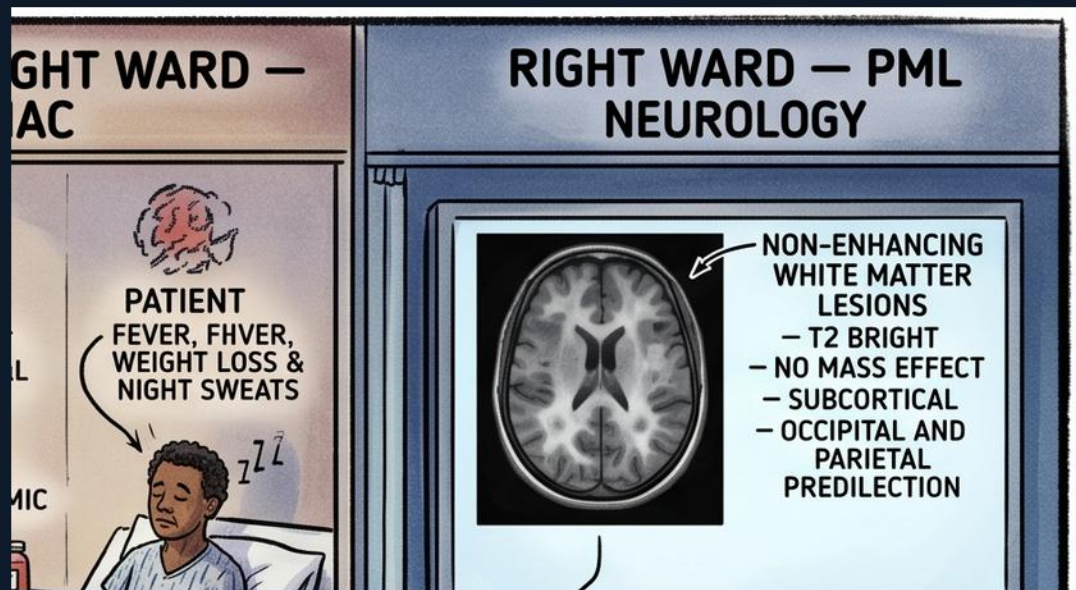
Clarithromycin + ethambutol ($\pm$ rifabutin) treatment shelf

WHAT IT ENCODES

Treat disseminated MAC with a MACROLIDE backbone (clarithromycin 500 mg BID or azithromycin 500–600 mg daily) PLUS ethambutol 15 mg/kg/day; add rifabutin as a third agent for severe disease, very low CD4, or high mycobacterial burden. Treat for ≥ 12 months and until immune reconstitution (CD4 > 100 for ≥ 6 months on ART). Watch ethambutol optic neuritis and rifabutin drug interactions/uveitis.

THE BOARD TRAP

WRONG: 'Clarithromycin (or azithromycin) monotherapy is fine for MAC.' Macrolide monotherapy rapidly selects resistance — always pair the macrolide with ethambutol ($\pm$ rifabutin). The discriminator is that combination therapy prevents macrolide resistance, which is what dooms single-agent treatment.



PML / JC virus

WHAT YOU SEE

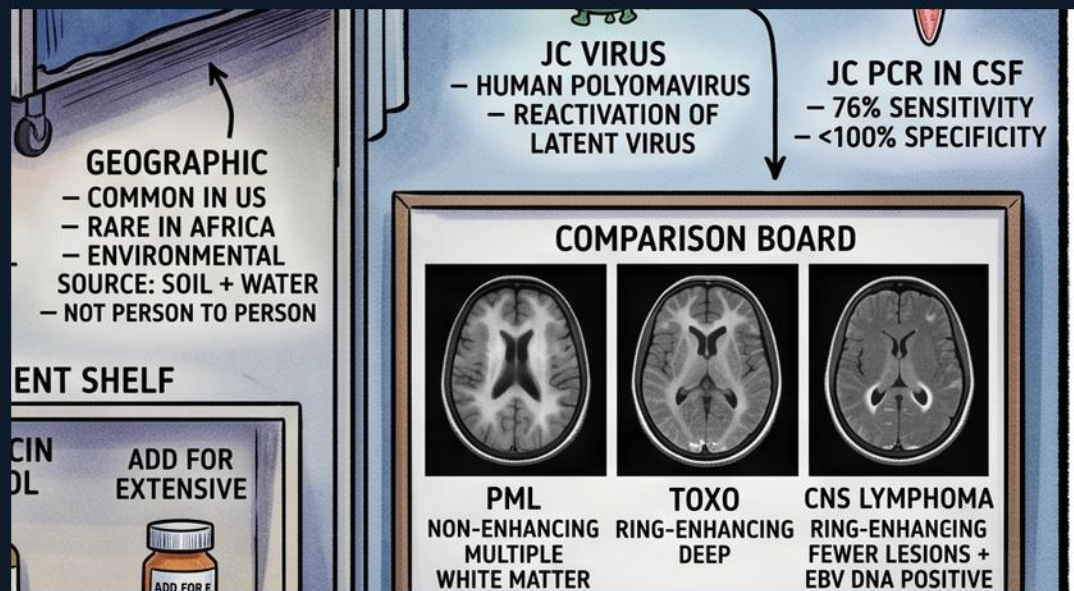
Right ward 'PML Neurology', brain MRI white matter lesions, JC virus

WHAT IT ENCODES

Progressive multifocal leukoencephalopathy (PML) is reactivation of JC polyomavirus in oligodendrocytes at low CD4 (<200, often <100). MRI shows multifocal, asymmetric, NON-enhancing white matter lesions with NO mass effect, typically subcortical/parieto-occipital. Clinically: subacute focal deficits — hemiparesis, visual field loss, cognitive/behavioral change, ataxia — without fever or headache.

THE BOARD TRAP

WRONG: 'Ring-enhancing brain lesions with mass effect = PML.' Ring-enhancing, mass-effect lesions point to toxoplasmosis (multiple) or CNS lymphoma (often single, EBV+, thallium/PET avid) — NOT PML. PML lesions are non-enhancing, lack mass effect, and don't respond to steroids; confusing the enhancement pattern is the classic distinguishing trap.



PML dx & treatment

WHAT YOU SEE

'JC PCR in CSF', treatment note: ART is the only effective therapy

WHAT IT ENCODES

Diagnose PML by detecting JC virus DNA on CSF PCR (high specificity) plus the characteristic MRI; brain biopsy is confirmatory if needed. There is NO specific antiviral for JC virus — the only effective treatment is immune reconstitution by starting/optimizing ART. Prognosis improves with ART, but IRIS can transiently worsen lesions and clinical status.

THE BOARD TRAP

WRONG: 'Treat PML with cidofovir/cytarabine (or steroids).' No antiviral cures PML; the definitive intervention is ART to restore immunity. Steroids are only used to manage severe PML-IRIS, not as primary therapy. Choosing an antiviral over ART is the trap — and clinicians must counsel that early ART-driven IRIS may cause transient worsening.

Summary — every symbol

1. PCP ward / CXR

Pneumocystis jirovecii pneumonia (PCP/PJP) classically shows diffuse bilateral perihilar interstitial (ground-glass) infiltrates, but the CXR can be normal early in disease

3. Mild PCP path

Mild-to-moderate PCP ($\text{PaO}_2 > 70$ mmHg AND A-a gradient < 35 mmHg): treat with TMP-SMX for 21 days, NO adjunctive steroids

5. TMP-SMX is first-line

TMP-SMX (trimethoprim-sulfamethoxazole) is first-line for both PCP treatment (21 days) and prophylaxis (one DS or SS tablet daily)

7. India ink / CrAg + ICP

Diagnosis: CSF/serum cryptococcal antigen (CrAg) is the most sensitive test; India ink shows encapsulated yeast in CSF

9. Ampho B + flucytosine

INDUCTION therapy = liposomal amphotericin B (3–4 mg/kg/day; or deoxycholate 0.7–1 mg/kg/day) PLUS flucytosine (5-FC, 100 mg/kg/day divided q6h) for ~2 weeks

11. MAC ward

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13. PML / JC virus

Progressive multifocal leukoencephalopathy (PML) is reactivation of JC polyomavirus in oligodendrocytes at low CD4 (< 200 , often < 100)

2. ABG / A-a gradient fork

Severity in PCP is stratified by room-air ABG: calculate PaO_2 and the alveolar-arterial (A-a) oxygen gradient to decide whether adjunctive corticosteroids are indicated

4. Severe PCP + steroids

Severe PCP ($\text{PaO}_2 < 70$ mmHg OR A-a gradient > 35 mmHg): add corticosteroids, started BEFORE or WITH the first antibiotic dose to blunt the inflammatory burst from dying organisms

6. Cryptococcal meningitis

Cryptococcal meningitis (Cryptococcus neoformans) presents subacutely over days to weeks with headache, fever, malaise, and often subtle meningismus; it occurs typically at $\text{CD4} < 100$

8. 3-phase tx ladder

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10. Fluconazole maintenance

After ~2 weeks of induction (and documented clinical improvement/CSF sterilization), step down to fluconazole: 400–800 mg/day for ~8 weeks (consolidation), then 200 mg/day (maintenance/secondary prophylaxis)

12. MAC treatment combo

Treat disseminated MAC with a MACROLIDE backbone (clarithromycin 500 mg BID or azithromycin 500–600 mg daily) PLUS ethambutol 15 mg/kg/day; add rifabutin as a third agent for severe disease, very low CD4 , or high mycobacterial burden

14. PML dx & treatment

Diagnose PML by detecting JC virus DNA on CSF PCR (high specificity) plus the characteristic MRI; brain biopsy is confirmatory if needed